

Week 14 Assessment – Azure VPN

Step 1: Create Resource Group

Home > Resource Manager | Resource groups

Create a resource group ...

Basics

Tags

Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ	Azure for Students
Resource group name * ⓘ	rg-vnet-vpn-lab
Region * ⓘ	(Asia Pacific) Central India

Home > Resource Manager

Resource Manager | Resource groups

Export resource groups using Bicep or Terraform

How to manage changes with deployment tools?

Summarize my costs by service

×

«

+ Create

⚙️ Manage view

↻ Refresh

↓ Export to CSV

🔗 Open query

|

🏷️ Assign tags

+

Add to service group

Group by none ▾

Resource Manager

All resources

★ Favorite resources

🕒 Recent resources

Resource groups

🏷️ Tags

> Organization

> Tools

> Deployments

> Help

Filter for any field...

Subscription equals all

Location equals all

✕

+ Add filter

☐	Name ↑		Subscription	Location
☐	rg-vnet-vpn-lab	...	Azure for Students	Central India
☐	VisualStudioOnline-5A2E30AE5F2B4608B2EBC834D76453E5	...	Azure for Students	South India

Add or remove favorites by pressing Ctrl+Shift+F

Showing 1 - 2 of 2. Display count: auto ▾

Give feedback

Step 2: Create VNet-1

Create virtual network ...

Basics Security Address space Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *

[Create new](#)

Instance details

Virtual network name *

Region * ⓘ

[Deploy to an Azure Extended Zone](#)

Step 3: Create VNet-2

Microsoft Azure

Search resources, services, and docs (G+)

Home > Network foundation | Virtual networks

Create virtual network ...

Basics Security Address space Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *

[Create new](#)

Instance details

Virtual network name *

Region * ⓘ

[Deploy to an Azure Extended Zone](#)

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Network foundation

Network foundation | Virtual networks

Preview

Check health of virtual networks | What new Azure preview features are there for networking? | Check the status of all virtual network routes

Search

+ Create | Manage view | Refresh | Export to CSV | Open query | Assign tags | Add to service group | Group by none

Filter for any field... | Subscription equals all | Resource Group equals all | Location equals all | Add filter

	Name ↑	Resource Group	Location	Subscription
☐	vnet1	rg-vnet-vpn-lab	Central India	Azure for Students
☐	vnet2	rg-vnet-vpn-lab	Central India	Azure for Students

Virtual network overview

Virtual networks

NAT gateways

Public IP addresses

Subnets for vnet1:

vnet1 | Subnets

Virtual network

Search

+ Subnet | Refresh | Manage users | Delete | Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Se
☐	default	10.0.0.0/24	-	251	-	-
☐	subnet1	10.0.1.0/24	-	251	-	-
☐	GatewaySu...	10.0.255.0/27	-	availability ...	-	-

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets for vnet2:

Home > vnet2-1775881743810 | Overview > vnet2

vnet2 | Subnets

Virtual network

Search

+ Subnet | Refresh | Manage users | Delete

Create subnets to segment the virtual network address space into sm

Search subnets

	Name ↑	IPv4
☐	default	10.1.0.0/24

Showing 1 subnet

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose

Name *

IPv4

Include an IPv4 address space ☒

IPv4 address range
10.1.0.0 - 10.1.255.255

Starting address *
10.1.1.0

Size
/24 (256 addresses)

Subnet address range
10.1.1.0 - 10.1.1.255

IPv6

Include an IPv6 address space ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Add Cancel

Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS

Add or remove favorites by pressing Ctrl+Shift+F

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose: Virtual Network Gateway

Name: GatewaySubnet

IPv4

Include an IPv4 address space: ☒

IP address range: 10.1.0.0/16

Starting address: 10.1.255.0

Size: /27 (32 addresses)

Subnet address range: 10.1.255.0 - 10.1.255.31

IPv6

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Add **Cancel**

Subnets

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
default	10.1.0.0/24	-	251	-	-	-
subnet1	10.1.1.0/24	-	251	-	-	-
GatewaySubnet	10.1.255.0/27	-	availability ...	-	-	-

Step 4: Create VPN Gateway for VNet-1

Create virtual network gateway

Basics Tags Review + create

Azure has provided a planning and design guide to help you configure the various VPN gateway options. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Azure for Students

Resource group: rg-vpn-lab (derived from virtual network's resource group)

Instance details

Name: vnet1-gateway

Region: Central India

Gateway Configuration

Gateway type: VPN

SKU: Basic

Generation: Generation1

Review + create **Previous** **Next: Tags** [Download a template for automation](#)

Step 5: Create VPN Gateway for VNet-2

Microsoft Azure

Search resources, services, and docs (G+)

Home

>

Hybrid connectivity | ExpressRoute gateways

Create virtual network gateway

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource group ⓘ

NetworkWatcherRG (derived from virtual network's resource group)

Instance details

Name *

vnet2-gateway

Region *

Central India

[Deploy to an Azure Extended Zone](#)

Gateway Configuration

Gateway type * ⓘ

☒ VPN

☐ ExpressRoute

SKU * ⓘ

Basic

Generation ⓘ

Generation1

Enable Advanced Connectivity ⓘ

☐ Enabled

☒ Disabled

Virtual network * ⓘ

vnet2

Microsoft Azure

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Home

>

Hybrid connectivity

Hybrid connectivity | VPN gateways

Check health of virtual network gateways

Analyze connectivity for listed virtual network gateways

How to ensure fault tolerance of gateways

Search

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Gateway type equals Vpn

Add filter

Overview

ExpressRoute

VPN gateway

Set up VPN Gateway

VPN gateways

VPN connections

Local network gateways

Virtual WAN

	Name ↑	Virtual network	Gateway type	Resource Group	Location	Subscription	Resource type
☐	vnet1-gateway	vnet1	Vpn	rg-vnet-vpn-lab	Central India	Azure for Students	Microsoft.Network/...
☐	vnet2-gateway	vnet2	Vpn	rg-vnet-vpn-lab	Central India	Azure for Students	Microsoft.Network/...

Step 6: Create Connection (VNet1 → VNet2)

Microsoft Azure

Search resources, services, and docs (G+/J)

Home > Hybrid connectivity | VPN gateways > vnet1-gateway | Connections

Create connection

BasicsSettingsTagsReview + create

Create a secure connection to your virtual network by using VPN Gateway or ExpressRoute.
[Learn more about VPN Gateway](#) [Learn more about ExpressRoute](#)

Project details

Subscription *

Azure for Students

Resource group *

rg-vnet-vpn-lab

Create new

Instance details

Connection type *

VNet-to-VNet

Establish bidirectional connectivity

Name *

vnet1-to-vnet2

Region *

Central India

Previous

Next : Settings >

Step 7: Create Reverse Connection (VNet2 → VNet1)

Microsoft Azure

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Home

CreateConnection-20260411114110 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateConnection-20260411114110
Subscription : Azure for Students
Resource group : rg-vnet-vpn-lab

Start time : 11/4/2026, 11:45:47 am
Correlation ID : be8ae8b4-1ded-44e2-882a-f1125019e587

Deployment details

Resource	Type	Status	Operation details
✓ vnet2-to-vnet1	✓ Connection	OK	Operation details
✓ vnet1-to-vnet2	✓ Connection	OK	Operation details

Next steps

Go to resource group

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud

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Step 8: Deploy Test VMs

VM1 in VNet1

Microsoft Azure

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Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload Help me create a low cost VM Help me create a VM optimized for high availability

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * rg-vnet-vpn-lab
Create new

Instance details

Virtual machine name * vm1

Region * (Asia Pacific) Central India
Deploy to an Azure Extended Zone

Availability options No infrastructure redundancy required

Security type Trusted launch virtual machines
Configure security features

< Previous Next: Disks > Review + create

Give feedback

VM2 in VNet2

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Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload Help me create a VM optimized for high availability Help me create a low cost VM

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * rg-vnet-vpn-lab
Create new

Instance details

Virtual machine name * vm2

Region * (Asia Pacific) Central India
Deploy to an Azure Extended Zone

Availability options No infrastructure redundancy required

Security type Trusted launch virtual machines
Configure security features

Image * Ubuntu Server 24.04 LTS - x64 Gen2
See all images | Configure VM generation

< Previous Next: Disks > Review + create

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Home > Compute infrastructure

Compute infrastructure | Virtual machines

Identify VMs with network connectivity issues List VMs by security posture Get workload templates for VMs

Search

Overview

All resources

Infrastructure

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

> Disks + images

Create Reservations Manage view Refresh Export to CSV Open query Assign tags Start Restart Stop Group by none

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
vm1	Azure for Stude...	rg-vnet-vpn-lab	Central India	Running	Linux	Standard_DC1s...	-	1
vm2	Azure for Stude...	rg-vnet-vpn-lab	Central India	Running	Linux	Standard_DC1s...	-	1

Step 9: Test Connectivity

The programs included with the Ubuntu system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "sudo <command>". See "man sudo_root" for details.

```
azureuser@vm1:~$ ping 10.1.1.4
PING 10.1.1.4 (10.1.1.4) 56(84) bytes of data.
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=5.37 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=4.37 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=5.50 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=3.81 ms
64 bytes from 10.1.1.4: icmp_seq=5 ttl=64 time=28.6 ms
64 bytes from 10.1.1.4: icmp_seq=6 ttl=64 time=3.42 ms
64 bytes from 10.1.1.4: icmp_seq=7 ttl=64 time=72.0 ms
64 bytes from 10.1.1.4: icmp_seq=8 ttl=64 time=3.63 ms
64 bytes from 10.1.1.4: icmp_seq=9 ttl=64 time=10.9 ms
64 bytes from 10.1.1.4: icmp_seq=10 ttl=64 time=3.43 ms
64 bytes from 10.1.1.4: icmp_seq=11 ttl=64 time=3.84 ms
^C
--- 10.1.1.4 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10016ms
rtt min/avg/max/mdev = 3.418/13.165/72.006/19.902 ms
azureuser@vm1:~$
```